ARTICLE

Peer Social Genetic Effects and the Etiology of Substance Use Disorders, Major Depression, and Anxiety Disorder in a Swedish National Sample

Jessica E. Salvatore, Ph.D., Henrik Ohlsson, Ph.D., Jan Sundquist, M.D., Ph.D., Kristina Sundquist, M.D., Ph.D., Kenneth S. Kendler, M.D.

Objective: There is growing interest in how peers' genotypes may influence health (i.e., peer social genetic effects). The authors sought to clarify the nature of peer social genetic effects on risk for drug use disorder, alcohol use disorder (AUD), major depression, and anxiety disorder.

Method: Cox models were used with data from a population-based Swedish cohort (N=655,327). Outcomes were drug use disorder, AUD, major depression, and anxiety disorder registrations between ages 17 and 30 from medical, criminal, and pharmacy registries. The authors indexed peer social genetic effects with peers' family genetic risk scores (FGRSs) for the same disorders, which are personalized measures of genetic risk inferred from diagnoses in first- to fifth-degree relatives.

Results: Across disorders, peer FGRSs predicted increased risks of proband registration (hazard ratio range, 1.01–1.59), with stronger effects for drug use disorder and AUD than for major depression and anxiety disorder. Peer social genetic effects were stronger for school classmates than for geographically proximal peers, and for peers from upper secondary school (ages 16–19) versus peers from lower secondary school (ages 7–16). Peer social genetic effects remained significant following statistical control for sociodemographic confounders, whether peers were affected, and peers' FGRS for educational attainment. Peer social genetic effects were more pronounced for probands at higher genetic risk.

Conclusions: The genetic makeup of adolescents' peers has long-reaching consequences on risks for drug use disorder, AUD, major depression, and anxiety disorder. Individuals at high genetic risk are more sensitive to social genetic effects. Alternative hypotheses such as sociodemographic stratification, exposure to affected peers, and genetic predispositions for educational attainment did not explain the risk associated with peer social genetic effects for substance use and psychiatric disorders.

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Genetic predispositions play a critical role in the etiology of common psychiatric conditions, including drug and alcohol use disorders, major depression, and anxiety disorder (1–4). Beyond the risk associated with one's own genotype, emerging evidence suggests that peers' genetic makeup may also contribute to substance use and psychiatric outcomes through what are known as social genetic effects (5–7). For example, classmates' and friends' polygenic predispositions to smoke predicted adolescents' smoking above and beyond the adolescents' own polygenic predispositions (5). Evidence for peer social genetic effects builds on a broader literature documenting social network influences on health (8) and, in particular, adolescent peer reinforcement for a range of outcomes, including behavior problems and depression (9, 10).

Several questions remain about the nature of peer social genetic effects for common substance use and psychiatric disorders. There is heterogeneity in how researchers can define peers and the developmental salience of peers increases across middle childhood and adolescence (11, 12); yet, it remains unclear whether the magnitude of social genetic effects changes across different definitions of peers (e.g., geographically proximal peers vs. school classmates), and whether there is developmental specificity in peer social genetic effects. Genetic predispositions for psychiatric disorders are stratified along socioeconomic dimensions (13, 14), but the robustness of peer social genetic effects to sociodemographic confounders remains unexplored. Relatedly, the degree to which peers' psychiatric diagnoses mediate social genetic effects or whether peers' predispositions for genetically correlated traits, such as educational attainment (15–19), explain social genetic effects for psychiatric disorders is not well understood. Finally, although

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the risk associated with peer social genetic effects might plausibly depend on individuals' own genetic predispositions (20), this possibility has not been formally tested.

We used Swedish national registry data (N=655,327) to characterize peer social genetic effects on probands' risk for registration for drug use disorder, alcohol use disorder (AUD), major depression, and anxiety disorder between ages 17 and 30. We indexed peers' genetic predispositions using family genetic risk scores (FGRSs), which are personalized measures of genetic risk inferred from diagnoses in first- to fifth-degree relatives (21). We asked the following questions:

1. Do peers' genetic predispositions for drug use disorder, AUD, major depression, or anxiety disorder predict proband registration for a disorder between ages 17 and 30? Does the strength of peer social genetic effects differ for geographically defined and school-based peers, does it differ for lower versus upper secondary school peers, and does it differ across disorders?
2. Are peer social genetic effects robust to sociodemographic and genetic confounders?
3. Do peers' substance use and psychiatric disorders mediate social genetic effects, and are peer social genetic effects robust to peer genetic predispositions for educational attainment?
4. Are individuals with higher genetic predispositions more sensitive to peer social genetic effects, such that they are more likely to develop drug use disorder, AUD, major depression, or anxiety disorder if exposed to genetically predisposed peers?

To evaluate potential bias, we also conducted a negative control analysis for height, an outcome for which we expect little to no peer social genetic effects.

METHODS

Swedish National Registries

We collected information on individuals from Swedish population-based registers that was linked to each person's unique personal identification number (replaced with a pseudonymized serial number by Statistics Sweden for confidentiality). Ethical approval was granted by the regional ethical review board in Lund, Sweden (no. 2008/409 and later amendments). Participant consent was not required. The database consisted of all individuals born in Sweden to Swedish-born parents from 1980 to 1998 followed through December 31, 2018 (N=1,566,985). In keeping with our goal of examining whether the strength of peer social genetic effects differs for geographically defined and school-based peers, we used registry information to map individuals by location and by school. For geographically defined peers, we used individuals' addresses (which, for confidentiality, were delivered to us in 250×250-meter coordinate squares) at the end of the year they turned 15. From this information, we assigned all individuals into three geographical areas: a 1,000-meter square, a demographic statistical area (i.e., areas whose boundaries take into account geographical features such as streets and waterways), and a municipality. For school-based peers, we used information from two national school registers to assign all individuals to their lower secondary school (approximately age 16) and their choice of programs or branches in upper secondary school (approximately age 18–19, depending on whether they were in a 2- or 3-year program). Hence, an individual belonged to six different geographical and administrative levels that were partly hierarchical (see Figure S1 in the online supplement for a schematic). We coded each geographical or administrative level independently per year of birth (e.g., individuals in the same lower secondary school were clustered only with individuals with the same year of birth) to identify peers belonging to the same geographical or administrative level. In Sweden, lower secondary school is compulsory, and upper secondary school is voluntary. For 231,943 individuals, we had no information on upper secondary school. This meant that we could investigate 85.1% (N=1,335,042) of the original sample. From previous studies, we know that individuals who did not start upper secondary school have higher rates of substance use and psychiatric disorders (22).

Measures

Our outcome measures were proband registration for drug use disorder, AUD, major depression, and anxiety disorder between ages 17 and 30. Definitions for these disorders across medical, pharmacy, and legal registries are provided in Table S1 in the online supplement. Our predictors of interest were peer FGRSs for these same disorders. The FGRSs were calculated from morbidity risks in first- through fifth-degree relatives, controlling for cohabitation effects, and thus arise from phenotypes in extended pedigrees (see Table S2 in the online supplement). Calculation of peer FGRSs proceeded in a series of steps that are detailed in the online supplement. Briefly, we employed a split-half, cross-classified multilevel linear regression model (to account for the nesting of individuals within their unique combinations of geographical and school-based classifications) with individual FGRS as the outcome variable. From this, we used the residuals (i.e., the deviation from the mean FGRS) at each level of analysis to quantify peer social genetic effects in the second split-half sample. These residuals were divided into five equally sized groups, and each individual in the second random half was categorized into six geographical and school levels that in turn were defined as high, mid-high, mid, mid-low, or low peer FGRS. This was done for each disorder separately.

By using the residuals from the cross-classified model, units with limited information (i.e., few individuals or large variation in FGRSs) were centered toward the mean value, and the residual was close to 0. This is more relevant for the 1,000-meter squares and demographic and statistical areas, where there were fewer individuals within each classification (see Table S3 in the online supplement). Using

municipality as an example, in the cross-classified models, there is an overall mean FGRS for all municipalities, and the residuals represent the deviation for each municipality from the overall mean. Thus, an individual living in a municipality with a large and positive FGRS residual for AUD would be classified as having high peer genetic predispositions to AUD. This approach is conceptually similar to other split-half or discovery-validation approaches where information from an independent sample informs definitions or calculations in the validation sample to reduce capitalizing on chance. Importantly, this approach also ensured that one's own FGRS did not contribute to the definition of peer FGRS. Note that in twin pairs, we randomly selected one twin, so that peer social genetic effects did not include information from close relatives. However, half-siblings and cousins that were born in the same year and belong to the same classification could have been included in the peer-effect calculations.

Statistical Analysis

To investigate evidence for peer social genetic effects, we used Cox proportional hazards models on the second split-half sample, in which all individuals had been assigned peer FGRSs based on residuals calculated from the first split-half sample. Using AUD as an example, we investigated the risk of AUD as a function of peer FGRS for AUD at the 6 different levels, from age 17 until end of the follow-up (death, emigration, age 30, or December 31, 2018), while controlling for sex, FGRS for AUD in the proband, and AUD registration prior to age 17. We treated the exposure variables as five-level categorical variables, and in the tables, we present the hazard ratio (and 95% confidence intervals) for high versus low peer genetic predisposition.

To investigate whether sociodemographic and genetic confounders accounted for peer social genetic effects, we created a family socioeconomic status factor score composite that included unemployment status, disposable income, educational status, social welfare recipient status, and immigrant status of biological parents when individuals were age 15 (see online supplement). The factor scores, together with an FGRS for education, were included in the Cox regression models (model B), and we investigated decreases in the exposure variables of interest (i.e., peer FGRSs). We also included a dummy variable defining whether the upper secondary school was the vocational or college-preparatory track.

To investigate whether peers' substance use and psychiatric disorder registrations mediated social genetic effects, and whether peer genetic predispositions for educational attainment confounded peer social genetic effects for substance use and psychiatric disorders, we replicated the cross-classified regression approach described above to create indices of peer drug use disorder, AUD, major depression, and anxiety disorder. Individuals' substance use and psychiatric disorder registrations prior to age 19 were included. We included these variables in the Cox regression model (model C) and investigated decreases in the exposure variables. Peer genetic predispositions for educational attainment were based on a cross-classified linear regression model and were included in the Cox regression model (model D).

To investigate whether proband genetic makeup modified peer social genetic effects such that probands with higher individual genetic predispositions were more likely to develop drug use disorder, AUD, major depression, or anxiety disorder if exposed to peers with high genetic predispositions, we modeled the interaction between peer FGRSs and the proband FGRSs. To focus on interactions on an additive scale, we used a linear probability model. Based on results from previous models, we investigated the interaction at the level with the highest peer social genetic effects (i.e., the program or branch type in upper secondary school). We used the same covariates as described for model C. We present the extra effect (measured as an increase in rates of the outcome), when comparing high versus low peer FGRS at different levels of proband FGRS.

Negative control analyses for height were run as linear regression models in a subsample of males (N=122,856) for whom height data were available in the Military Conscription Register (see online supplement).

All statistical analyses were performed using SAS, version 9.4, and MLwiN, version 3.02.

RESULTS

Descriptive Statistics

Table 1 summarizes key characteristics of the study cohort.

Peer Social Genetic Effects

In our first Cox proportional hazard model, peer FGRSs across geographical and school-based definitions predicted increased risk of proband registration, with generally stronger associations for school-based peers (hazard ratio range, 1.03–1.59) than for geographically defined peers (hazard ratio range, 1.01–1.09) (Table 2, model A). There was also evidence of developmental specificity. Peer social genetic effects were strongest within type of upper secondary school (i.e., classmates in the same track [either vocational or college-preparatory track] between ages 16 and 19), followed by upper secondary school (i.e., all classmates regardless of track), and then lower secondary school (i.e., classmates between ages 7 and 16). Social genetic effects for school-based peers were stronger for drug use disorder and AUD compared with anxiety disorder and major depression. In contrast, the strength of social genetic effects of geographically defined peers did not differ across disorders.

Robustness to Sociodemographic and Genetic Confounders

In our second Cox proportional hazard model, the hazard ratios for the peer FGRSs changed only modestly with the

TABLE 1. Descriptive statistics for the 655,327 individuals (322,549 female and 332,778 male) included in the analysis of peer social genetic effects on substance use and psychiatric disorders[a]

Characteristic	Disorder															
	MD				AD				DUD				AUD			
	%				%				%				%			
Overall rate	12.3				14.8				3.9				1.9			
Female	16.2				19.8				2.0				1.3			
Male	8.5				9.9				5.8				2.4			
	Mean		SD		Mean		SD		Mean		SD		Mean		SD	
Age at registration	23.6		3.6		23.7		3.5		22.1		3.4		22.8		3.4	
Female	23.5		3.6		23.5		3.5		22.0		3.2		22.6		3.4	
Male	23.8		3.5		24.0		3.4		22.0		3.3		23.0		3.4	
	MD		No MD		AD		No AD		DUD		No DUD		AUD		No AUD	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
FGRS	0.23	1.0	−0.12	0.9	0.20	1.0	−0.13	0.9	0.31	1.2	−0.12	0.8	0.34	1.1	−0.10	0.9
FGRS for education	0.02	1.0	0.06	1.0	0.02	1.0	0.06	1.0	−0.14	0.9	0.06	1.0	−0.19	0.9	0.06	1.0
Low SES score	0.03	0.9	−0.08	0.8	0.02	0.9	−0.08	0.7	0.16	1.1	−0.07	0.8	0.15	1.1	−0.07	0.8
	%		%		%		%		%		%		%		%	
Registration prior to age 17	1.6		0.2		1.4		0.2		2.2		0.2		1.5		0.1	
Peers in top FGRS quintile																
Municipality	30.2		27.4		32.6		29.0		37.7		32.6		33.0		29.4	
DESO	22.1		21.0		22.4		21.2		21.1		19.6		21.6		19.6	
1000-meter square	24.3		22.5		25.6		23.2		27.6		22.6		25.4		22.6	
Upper secondary school	29.0		25.2		29.2		25.6		29.3		21.3		29.3		22.3	
Upper secondary school program type	28.2		24.5		28.9		24.2		29.1		20.6		28.2		20.7	
Lower secondary school	24.9		23.3		25.0		23.2		24.6		22.0		24.6		22.2	

[a] All individuals belong to six different classifications. Three classifications are geographically defined (municipality, DESO, and a 1,000-meter square), and three are school-based (upper secondary school, vocational vs. college-preparatory program type in upper secondary school, and lower secondary school). The sample (N=655,327) summarized here corresponds to the second split-half sample in which inferential analyses were conducted, following the cross-classified multilevel regression model calculation of peer social genetic effects in the first split-half sample. AD=anxiety disorder; AUD=alcohol use disorder; DESO=demographic statistical area; DUD=drug use disorder; FGRS=family genetic risk score; MD=major depression; SES=socioeconomic status.

inclusion of sociodemographic and genetic confounders (proband socioeconomic status, proband FGRS for educational attainment, and whether one was in a vocational or college-preparatory track in upper secondary school) (Table 2, model B). The exceptions to this general pattern were that there were more pronounced reductions in the hazard ratios for peer FGRSs for drug use disorder and for AUD at the upper secondary and type of upper secondary levels. For example, within the type of upper secondary school level, the hazard ratio for peer FGRS for drug use disorder went from 1.59 to 1.32, and the hazard ratio for peer FGRS for AUD went from 1.56 to 1.24.

Mediation of Peer Social Genetic Effects

In our third Cox proportional hazard model, inclusion of the corresponding peer substance use or psychiatric disorder phenotype reduced the magnitude of the hazard ratios for the peer FGRS only modestly (Table 2, model C; Figure 1) (the full results from this model are provided in Table S5 in the online supplement). Of note, the hazard ratios for the peer substance use and psychiatric disorders were larger for school-based peers than for geographically defined peers. Additionally, the hazard ratios for the peer FGRSs and the peer disorders were of similar magnitude and had largely overlapping confidence intervals. In our fourth Cox proportional hazard model, inclusion of peer FGRS for educational attainment as a covariate resulted in only a modest reduction in the hazard ratios for the peer FGRSs for the substance use and psychiatric disorders (Table 2, model D).

Genetic Sensitivity to Peer Social Genetic Effects

Peer social genetic effects were more pronounced among probands with higher FGRSs compared with probands with lower FGRSs (Figure 2). As a proband's FGRS increased from low (i.e., one standard deviation below the mean) to very high (i.e., two standard deviations above the mean), the prevalence for each disorder associated with high peer FGRS increased 1.47% for drug use disorder, 1.19% for AUD, 0.73% for major depression, and 0.88% for anxiety disorder.

Negative Control Analyses

Peer FGRSs for height within municipality and within upper secondary school were associated with proband height at a statistically significant level, but the effects were of little practical significance (see Table S6 in the online

TABLE 2. Cox regression models test of peer social genetic effects for drug use disorder, alcohol use disorder, major depression, and anxiety disorder across six geographical and school-based definitions of peers[a]

Geographical or School-Based Peer Classification Across Disorders	Model A		Model B		Model C		Model D	
	Hazard Ratio	95% CI	Hazard Ratio	95% CI	Hazard Ratio	95% CI	Hazard Ratio	95% CI
Drug use disorder								
Municipality	1.07	1.03, 1.11	1.10	1.06, 1.15	1.06	1.02, 1.10	1.06	1.02, 1.10
DESO	1.03	0.99, 1.07	1.01	0.97, 1.05	1.01	0.97, 1.05	1.02	0.97, 1.06
1,000-meter square	1.09	1.05, 1.13	1.06	1.03, 1.10	1.06	1.02, 1.10	1.07	1.03, 1.11
Lower secondary school	1.06	1.03, 1.10	1.05	1.01, 1.09	1.04	1.01, 1.08	1.07	1.03, 1.12
Upper secondary school	1.27	1.23, 1.32	1.17	1.12, 1.21	1.14	1.10, 1.18	1.13	1.09, 1.18
Upper secondary school (type)	1.59	1.53, 1.65	1.32	1.27, 1.37	1.28	1.24, 1.33	1.22	1.17, 1.26
Alcohol use disorder								
Municipality	1.01	0.96, 1.07	1.05	0.99, 1.11	1.05	0.99, 1.11	1.05	0.99, 1.11
DESO	1.05	0.99, 1.11	1.03	0.96, 1.08	1.03	0.97, 1.09	1.04	0.98, 1.10
1,000-meter square	1.05	0.99, 1.10	1.02	0.97, 1.08	1.02	0.97, 1.08	1.02	0.96, 1.08
Lower secondary school	1.07	1.02, 1.13	1.06	1.00, 1.11	1.05	1.00, 1.11	1.08	1.02, 1.14
Upper secondary school	1.22	1.16, 1.29	1.13	1.07, 1.19	1.12	1.06, 1.18	1.11	1.05, 1.18
Upper secondary school (type)	1.56	1.48, 1.65	1.24	1.17, 1.31	1.23	1.17, 1.31	1.15	1.08, 1.22
Anxiety disorder								
Municipality	1.02	1.00, 1.04	1.03	1.01, 1.05	1.03	1.00, 1.05	1.02	1.00, 1.04
DESO	1.02	1.00, 1.04	1.01	0.99, 1.04	1.01	0.99, 1.04	1.01	0.99, 1.03
1,000-meter square	1.03	1.01, 1.05	1.03	1.01, 1.05	1.03	1.00, 1.04	1.02	1.00, 1.04
Lower secondary school	1.03	1.01, 1.05	1.03	1.01, 1.05	1.03	1.01, 1.05	1.04	1.02, 1.06
Upper secondary school	1.08	1.06, 1.10	1.08	1.06, 1.10	1.06	1.04, 1.08	1.06	1.04, 1.08
Upper secondary school (type)	1.19	1.17, 1.21	1.17	1.15, 1.19	1.16	1.13, 1.18	1.14	1.12, 1.17
Major depression								
Municipality	1.02	1.00, 1.04	1.03	1.00, 1.05	1.02	1.00, 1.04	1.02	0.99, 1.04
DESO	1.02	1.00, 1.04	1.01	0.99, 1.04	1.01	0.99, 1.04	1.01	0.99, 1.04
1,000-meter square	1.03	1.00, 1.05	1.02	1.00, 1.04	1.02	1.00, 1.04	1.02	1.00, 1.04
Lower secondary school	1.04	1.02, 1.06	1.04	1.02, 1.06	1.04	1.02, 1.06	1.04	1.02, 1.07
Upper secondary school	1.11	1.08, 1.13	1.10	1.08, 1.12	1.08	1.06, 1.10	1.08	1.06, 1.10
Upper secondary school (type)	1.15	1.13, 1.18	1.13	1.10, 1.15	1.12	1.09, 1.14	1.11	1.08, 1.13

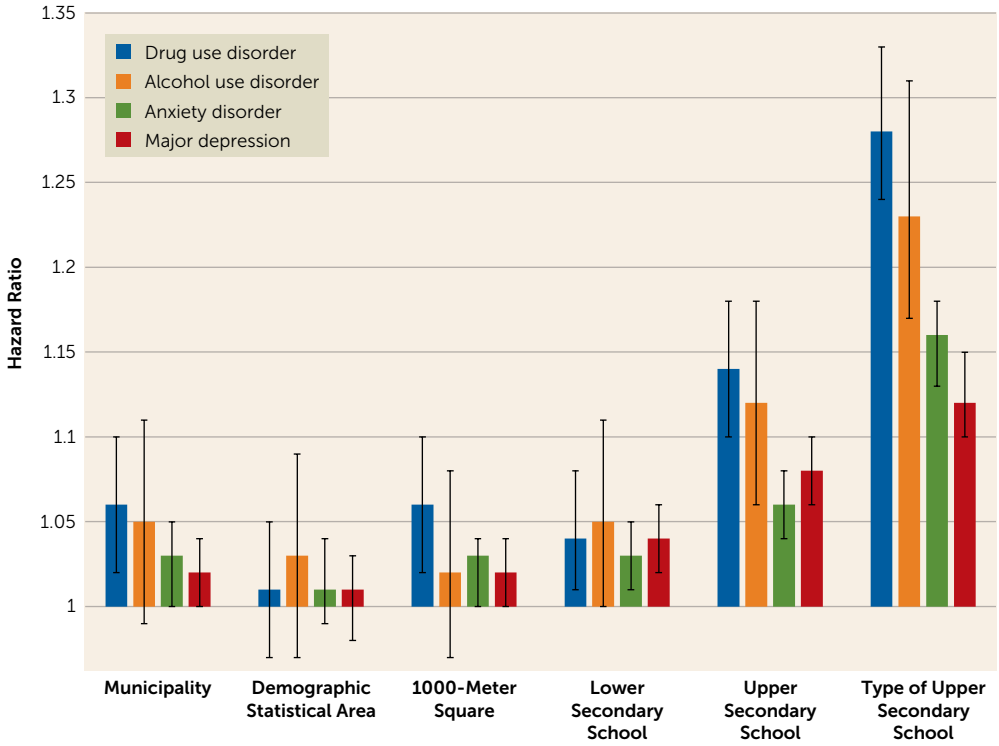
[a] Hazard ratios correspond to the association between the aggregated peer FGRS (highest quintile vs. lowest quintile) at the noted geographical or school-based classification and risk of proband registration for that disorder between ages 17 and 30. Models differed in the covariates included. Model A included aggregated peer FGRS, individual FGRS, sex, and individual registration at ages 15–16. Model B included aggregated peer FGRS, individual FGRS, sex, individual registration at ages 15–16, individual socioeconomic status, individual FGRS for educational attainment, and type of upper secondary school (vocational or college-preparatory track). Model C included aggregated peer FGRS, individual FGRS, sex, individual registration at ages 15–16, individual socioeconomic status, individual FGRS for educational attainment, type of upper secondary school (vocational or college-preparatory track), and aggregated peer registration for substance use and psychiatric disorders. Model D included aggregated peer FGRS, individual FGRS, sex, individual registration at ages 15–16, individual socioeconomic status, individual FGRS for educational attainment, type of upper secondary school (vocational or college-preparatory track), aggregated peer registration for substance use and psychiatric disorders, and aggregated peer FGRS for educational attainment. All individuals belong to six different classifications. Three classifications were geographically defined (municipality, DESO, 1,000-meter square), and three were school-based (lower secondary school, upper secondary school, and type of upper secondary school [vocational vs. college-preparatory track]). DESO=demographic and statistical area; FGRS=family genetic risk score.

supplement). For example, at the municipality level there was a 0.25-cm difference associated with peer FGRS for height in the top quintile. To more directly compare effect sizes across the primary substance use and psychiatric disorder outcomes of interest and the negative control analyses based on height, we also present the effect sizes (Cohen's d) corresponding to the peer social genetic effects in Table 3. Using the conventions of 0.2 (small), 0.5 (medium), and 0.8 (large) to characterize effect sizes, the peer social genetic effects within the type of upper secondary school (where the effects were strongest) were of very small magnitude for height, anxiety disorder, and major depression and of small magnitude for drug use disorder and alcohol use disorder. Thus, for the key analyses within type of upper secondary school, the Cohen's d statistic was over five times greater for drug use disorder and AUD than for that obtained for the negative control of height.

DISCUSSION

We sought to clarify the nature of peer social genetic effects in the etiology of common substance use and psychiatric disorders. Four key findings emerged. First, and consistent with previous studies (5, 6, 23–25), peer social genetic effects—that is, peers' genetic predispositions for drug use disorder, AUD, major depression, and anxiety disorder—prospectively predicted proband risk for substance use

FIGURE 1. Peer social genetic effects, capturing the association between the peer family genetic risk scores and proband substance use and psychiatric disorder outcomes[a]

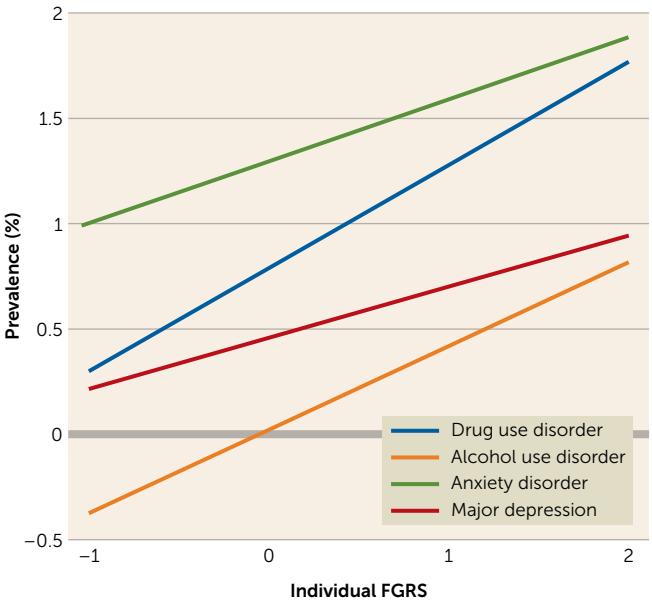


[a] Cox regression models predicting drug use disorder, alcohol use disorder, anxiety disorder, and major depression between ages 17 and 30. The hazard ratios are the risk in the top quintile versus the lowest. Results based on data provided in Table 2 (model C).

and psychiatric disorders. The effects were stronger for drug use disorder and AUD compared with anxiety disorder and major depression. Importantly, these peer social genetic effects held above and beyond the probands' own FGRS, which reduces concerns about individuals' own genetic predispositions as confounders in view of evidence that genetic predispositions for many traits and behaviors are geographically clustered (13). Previous studies of social genetic effects have largely relied on school-based definitions of peers (5) or clearly defined dyadic relationships (e.g., spouses [24, 26, 27] or friends [25]). Our study provides novel evidence for peer social genetic effects for an expanded definition of geographically defined peers. Still, peer social genetic effects for classmates in school were stronger than those for geographically defined peers, highlighting school class as a primary socializing unit and target for preventive interventions for the social contagion of substance use and psychiatric disorders. Our negative control analyses of height suggested there was minimal bias in our estimates of social genetic effects for these common psychiatric disorders.

Peer social genetic effects were also stronger for upper secondary school peers than for lower secondary school peers, which provides some support for the developmental specificity hypothesis. This result is consistent with the broader findings on the increasing significance of peers across adolescence (28) and underscores the long reach of late adolescent peers on subsequent risk for the onset of substance use and psychiatric disorders. Differences in the strength of the associations across disorders are also informative about the potential mechanisms underlying peer

FIGURE 2. Interaction between individuals' family genetic risk scores (FGRSs) and peers' FGRSs within the same type of upper secondary school[a]



[a] For each disorder, the change in prevalence (represented as a percent) associated with peer FGRS in the highest quintile versus the lowest quintile is plotted as a function of the proband's FGRS.

TABLE 3. Effect sizes (Cohen's d) corresponding to the peer social genetic effects for drug use disorder, alcohol use disorder, anxiety disorder, major depression, and height across six geographical and school-based definitions of peers[a]

Peer Social Genetic Effect	Drug Use Disorder	Alcohol Use Disorder	Anxiety Disorder	Major Depression	Height
Municipality	0.037	0.005	0.011	0.011	0.037
DESO	0.010	0.027	0.011	0.011	0.013
1,000-meter square	0.047	0.027	0.016	0.016	0.010
Lower secondary school	0.032	0.037	0.016	0.022	0.006
Upper secondary school	0.132	0.111	0.042	0.057	0.040
Upper secondary school (type)	0.256	0.245	0.047	0.077	0.047

[a] All individuals belong to six different classifications. Three classifications are geographically defined (municipality, DESO, 1,000-meter square) and three are school-based (lower secondary school, upper secondary school, and type of upper secondary school [vocational vs. college-preparatory track]). DESO=demographic and statistical area.

social genetic effects. School peers' FGRSs were stronger predictors of drug use disorder and AUD compared with anxiety disorder and major depression, which may be attributable to adolescents' selection into peer groups with similarly high genetic predispositions for drug use disorder and AUD, followed by the direct socialization and reinforcement of problem behaviors among peers through what is known as deviancy training (10, 29, 30). The relatively weaker associations observed for anxiety disorder and major depression mirrors the comparatively mixed literature about peer selection and socialization processes for internalizing problems, which may be limited to only certain types of relationships (30, 31). Our observation that social genetic effects for anxiety disorder and major depression were strongest for peers in the same type of upper secondary school (i.e., a relatively small socialization unit in which one may be likely to develop reciprocal friendships) is consistent with evidence that interpersonal processes such as co-rumination (32) and depressogenic attributional style (33) may contribute to the contagion of depressive symptoms in adolescent friendships.

Second, we found little support for the alternative explanation that variables associated with both exposure to peers with high genetic predispositions and probands' risk for registration for substance use and psychiatric disorders confound peer social genetic effects. Including measures of proband socioeconomic status, proband genetic predispositions for educational attainment, and whether one was in a vocational versus college-preparatory track in upper secondary school reduced peer social genetic effects for major depression and anxiety disorder only modestly. Reductions in peer social genetic effects were more pronounced for drug use disorder and AUD. Inclusion of the FGRS for educational attainment likely represents a quite conservative correction when testing for social genetic effects, given that genetic factors influencing substance use disorders are strongly correlated with genetic influences on educational attainment (15–18).

Third, we found minimal evidence to support one obvious explanation for the associations between peer FGRS and proband disorder, which is that peers with higher FGRSs are more likely to be affected, and in turn, direct exposure to affected peers drives contagion (i.e., peer FGRS increases the likelihood of peer disorder, which increases the likelihood of proband disorder). Instead, inclusion of peer diagnoses only modestly attenuated social genetic effects. This pattern implies that the mechanisms of peer social genetic effects need not be limited to the overt expression of the disorder in peers. In an earlier investigation, we similarly found that a spouse's genetic predispositions for alcohol or drug use disorders were associated with the partner's risk even if the spouse was unaffected (26, 27). Identifying the subclinical or premorbid traits and behaviors that mediate peer social genetic effects remains an important direction for future work. Given the robust evidence that many psychiatric disorders have a shared genetic architecture (34), whether cross-disorder diagnoses mediate social genetic effects also merits consideration. There was minimal evidence for the alternative explanation that peers' genetic predispositions for educational attainment confounded social genetic effects. As a set, these findings highlight the distinctiveness of peer social genetic effects in the etiology of substance use and psychiatric disorders.

Fourth, the risk associated with social genetic effects also depended on the proband's own genetic predispositions. Probands with higher genetic predispositions were at higher risk of developing a disorder if they were also exposed to peers with high genetic predispositions, although the magnitude of these effects was quite modest. This pattern of effects is consistent with the contextual triggering or diathesis-stress mechanisms of gene-by-environment interaction (35, 36), whereby an environmental exposure triggers an underlying predisposition. This finding also conceptually replicates previous evidence that peers are a particularly potent environmental moderator of genetic risk, especially for substance use and related externalizing outcomes (37–39). The implication of this finding is that individuals' own genetic makeup may render them more vulnerable to peer social genetic effects.

The findings should be interpreted in the context of several limitations. Analyses were limited to the Swedish population, and the potential generalizability to other populations is unknown. The FGRS approach depends on the accuracy of diagnoses and the performance of the cohabitation correction. Registry-based definitions tend to identify more severe cases. Encouragingly, the prevalence of registrations for these disorders in the Swedish national registries is quite consistent with the rates reported in a representative population sample of adults interviewed with the Composite

International Diagnostic Interview in nearby Oslo, Norway (40). Previous studies of the validity of the FGRS approach have demonstrated that these phenotype-based measures of quantitative genetic risk perform as expected (21, 41–49) and agree well with other similar statistical approaches (50). There is a high degree of concordance across our different methods of ascertainment for AUD and drug use disorder (51), as well as evidence for the validity of psychiatric diagnoses from primary care register data (52). Extensive simulations indicated that the correction for the shared environment performed as expected, suggesting that FGRSs largely capture genetic risk rather than shared environmental risk (49). Although registry-based data make it possible to use geographical and school-based definitions of peers, there is no further information about whether peers have close relationships with one another as friends or romantic partners. We note that in a previous investigation of peer social genetic effects for adolescent smoking behavior, the authors found no evidence that social genetic effects were stronger for friends than for grade-level peers (5). Finally, split-sample methods can be sensitive to sampling variation. In follow-up analyses, the observed effects were robust when a different sampling approach was used (see Table S7 in the online supplement).

CONCLUSIONS

Our findings in Swedish national data provide strong evidence that social genetic effects are an important etiological factor in the development of common substance use disorders and to a lesser extent major depression and anxiety disorder, above and beyond one's own genetic predispositions, well-established sociodemographic and genetic correlates, and exposure to affected peers. Individuals with high-risk genetic predispositions may be especially sensitive to the impact of social genetic influences. This speaks to the enduring significance of peer social genetic effects on the development of these common disorders, as well as the importance of considering peers as part of preventive intervention efforts.

AUTHOR AND ARTICLE INFORMATION

Department of Psychiatry, Robert Wood Johnson Medical School, Rutgers University (Salvatore); Center for Primary Health Care Research, Lund University, Malmö, Sweden (Ohlsson, Jan Sundquist, Kristina Sundquist); Virginia Institute for Psychiatric and Behavioral Genetics, Department of Psychiatry, and Department of Human and Molecular Genetics, Virginia Commonwealth University, Richmond (Kendler).

Send correspondence to Dr. Salvatore (jessica.salvatore@rutgers.edu) and Dr. Kendler (kenneth.kendler@vcuhealth.org).

Dr. Kristina Sundquist and Dr. Kendler share last authorship.

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Examination Questions for "Peer Social Genetic Effects and the Etiology of Substance Use Disorders, Major Depression, and Anxiety Disorder in a Swedish National Sample"

1. **What are peer social genetic effects?**
   A. Epigenetic changes caused by the social environment
   B. The influence of genes on social behavior
   C. The influence of a peer's genotype on another's behavior
   D. The effect of alleles that parents do not transmit to their offspring.
2. **Which of the following describes the relative strength of peer social genetic effects across psychiatric and substance use disorder diagnoses?**
   A. Peer social genetic effects are stronger for alcohol and drug use disorders compared to major depression and anxiety disorder.
   B. There is no evidence for peer social genetic effects.
   C. Peer social genetic effects are of the same magnitude across disorders.
   D. Peer social genetic effects are stronger for major depression and anxiety disorder compared to alcohol and drug use disorders.
3. **For whom are peer social genetic effects for psychiatric and substance use disorders stronger?**
   A. Individuals who are socially rejected
   B. Individuals who are genetically predisposed to these disorders
   C. Males
   D. Individuals with higher educational attainment